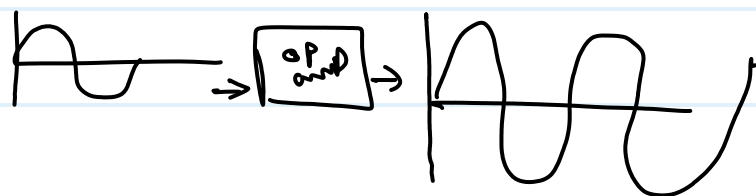


opamp



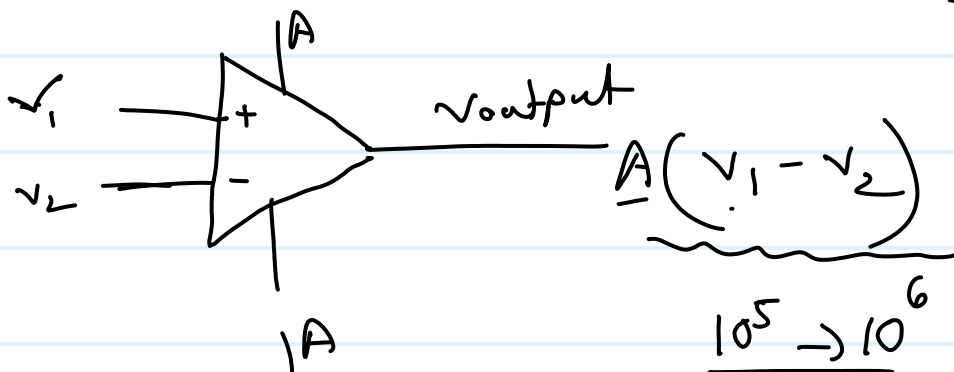
$$1V \rightarrow 10V$$

$$A \Rightarrow 10V$$

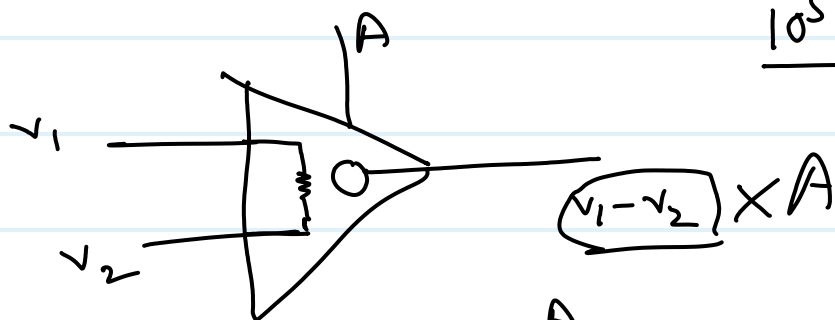
$$2V \Rightarrow$$

$$v_{out} \Rightarrow 10 \times 2 \\ \Rightarrow 20V$$

$$v_{out} \Rightarrow A \times v_{in}$$



$$10^5 \rightarrow 10^6$$

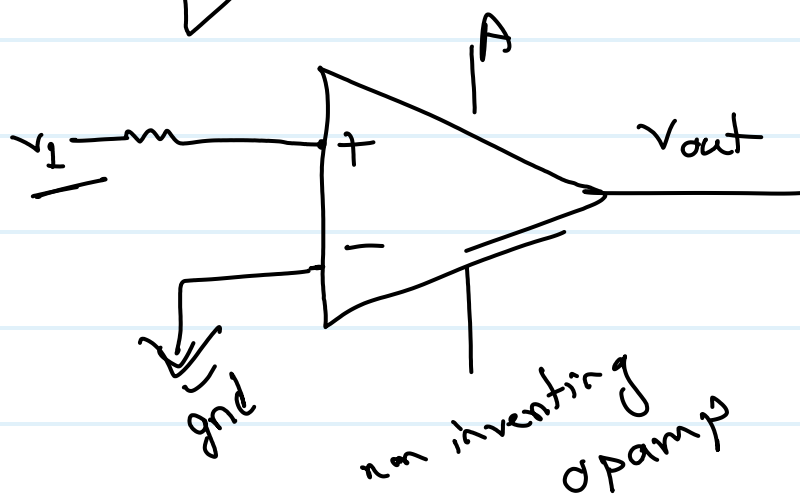


$$1V \rightarrow 10V$$

$$A \Rightarrow 10$$

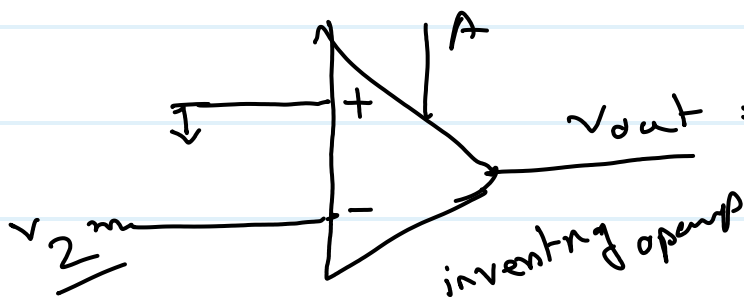
$$v_{in} \Rightarrow 1V$$

$$out \Rightarrow 10 \times 1 \\ \Rightarrow 10V$$



$$A(v_1 - v_2) \\ \Rightarrow A(v_1 - 0) \\ \Rightarrow +Av_1$$

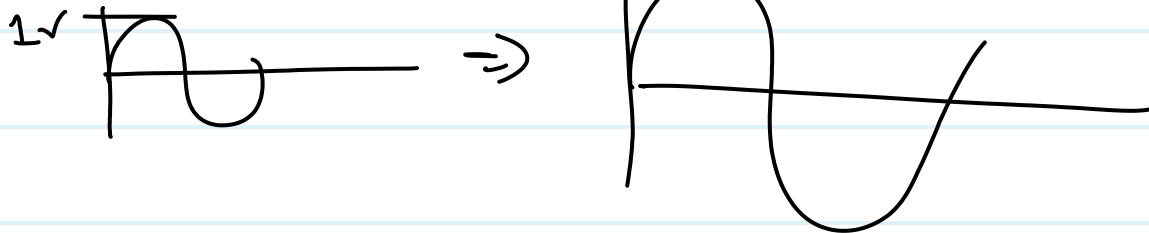
$$1V \rightarrow 10V$$



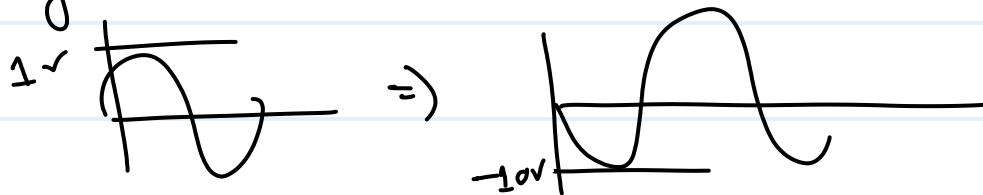
$$v_{out} \Rightarrow A(v_1 - v_2) \\ \Rightarrow A(0 - v_2) \\ \Rightarrow -Av_2$$

$$1V \rightarrow -10V$$

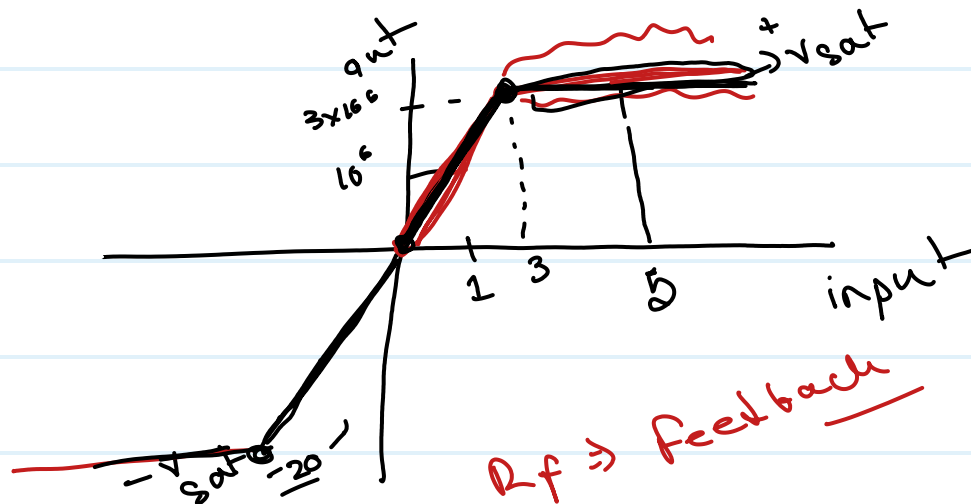
non inventory



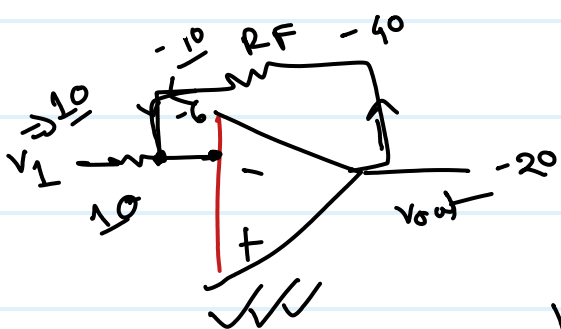
inventory



$1 \Rightarrow A \Rightarrow 10^5 - 10^6 \Rightarrow 1V \Rightarrow \text{output} \Rightarrow 1 \times 10^6 \checkmark$



$1 \Rightarrow 10^6$
 $2 \Rightarrow 2 \times 10^6$
 $3 \Rightarrow 3 \times 10^6$
 $4 \Rightarrow 3 \times 10^6$
 $5 \Rightarrow 3 \times 10^6$

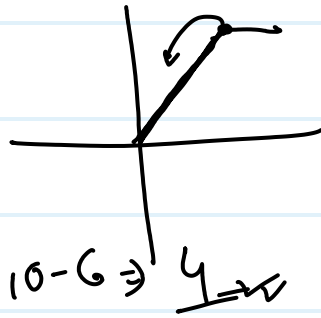


$10 \Rightarrow -\frac{20}{3}$
 $3 \Rightarrow \downarrow -9$

$\swarrow$ output $\uparrow$ feedback $\downarrow$
 output $\downarrow$ feedback $\uparrow$

input $\Rightarrow 20$, output $\Rightarrow -40$
 $20 - 10 \Rightarrow 10$

input $\Rightarrow 10$, output $\Rightarrow -20$
 $R_f \Rightarrow -7$
 $10 - 7 \Rightarrow 3$
 output $\Rightarrow -19$
 $R_f \Rightarrow -6$



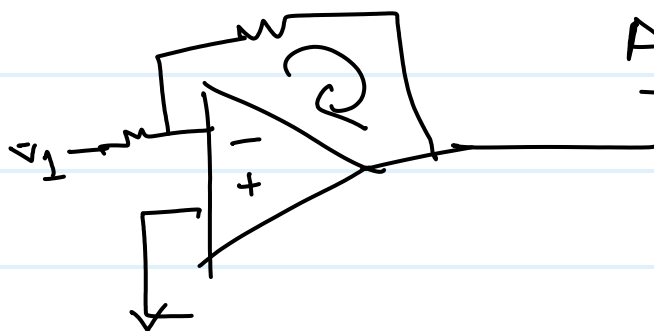
input $\Rightarrow 8 \rightarrow$ output $\Rightarrow -18$

$R_F \Rightarrow -5$

$10 \rightarrow 5 \Rightarrow 5 \rightarrow 4 \rightarrow 5$

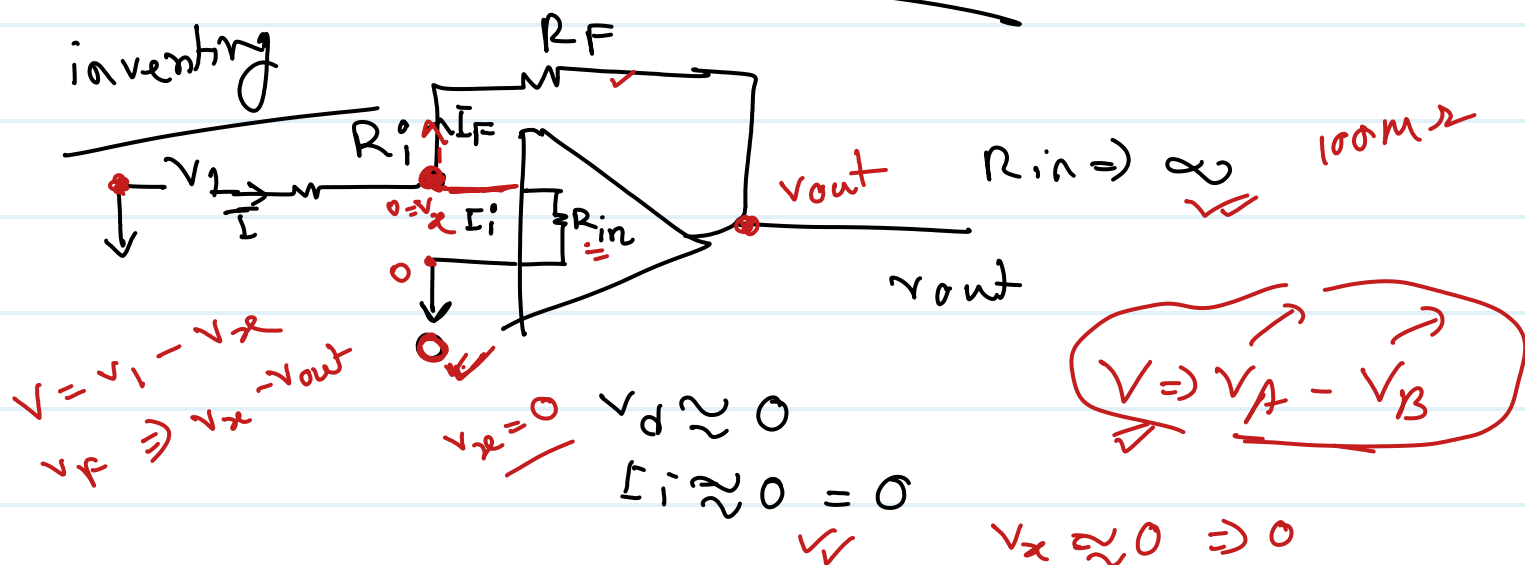


A_{CL} $\checkmark \checkmark$



ACL prove

inverting



$$I = I_F + I_i$$

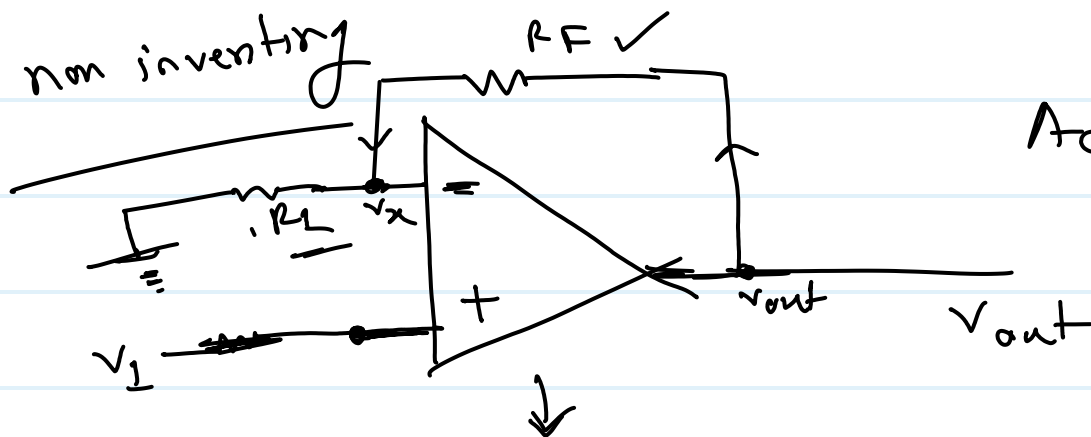
$$I \Rightarrow I_F$$

$$\Rightarrow \frac{V}{R} \Rightarrow \left(\frac{V_F}{R_F} \right) \Rightarrow \frac{V_1 - V_x}{R_1} = \frac{V_x - V_{out}}{R_F}$$

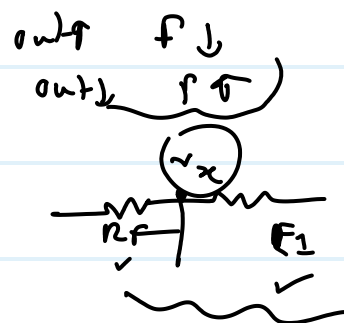
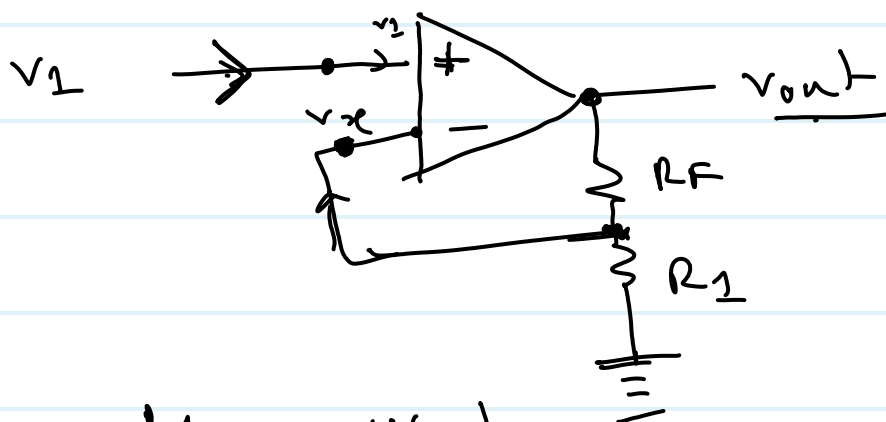
$$\Rightarrow \frac{V_1 - 0}{R_1} = \frac{0 - V_{out}}{R_F}$$

$$\frac{V_1}{R_1} = -\frac{V_{out}}{R_F}$$

inverting $\left\{ \frac{V_{out}}{V_{in}} = \frac{V_{out}}{V_1} = -\frac{R_F}{R_1} \Rightarrow A_{CL} \right.$



$$A_{CL} \Rightarrow \frac{V_{out}}{V_{in}} \Rightarrow ?$$



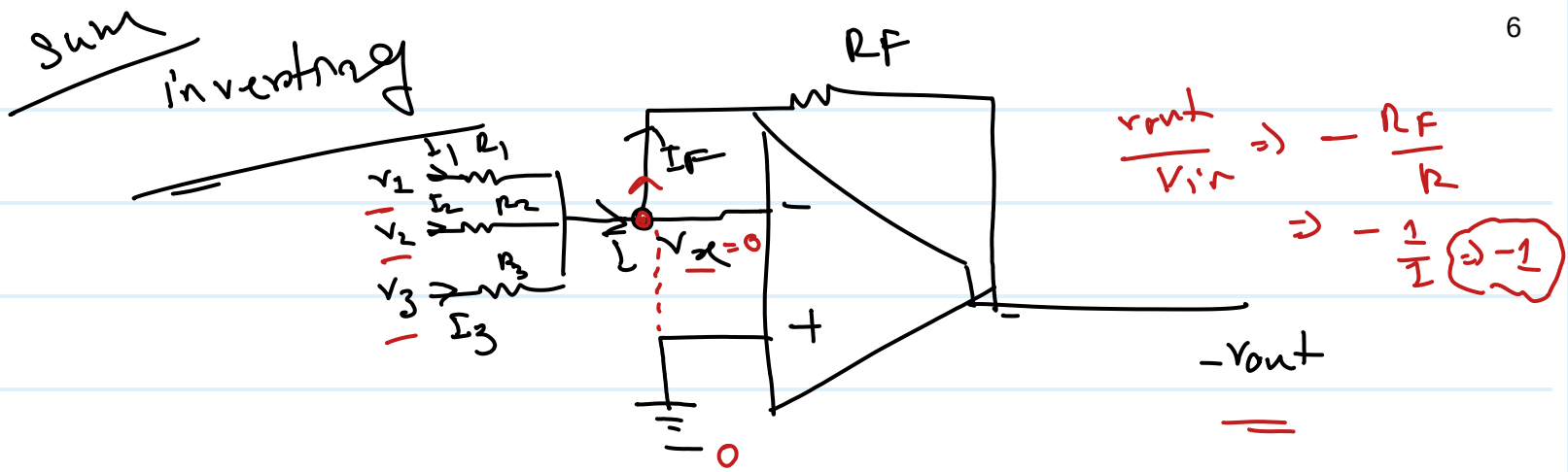
$$V_x \Rightarrow \frac{R_1}{R_F + R_1} \times V_{out}$$

$$V_x \Rightarrow \frac{R_F}{R_F + R_1} \times V_{out}$$

$$V_x \Rightarrow V_1$$

$$V_1 \Rightarrow \frac{R_1}{R_F + R_1} \times V_{out}$$

$$\frac{V_{out}}{V_1} \Rightarrow 1 + \frac{R_F}{R_1} \Rightarrow A_{CL}$$



$$I_1 + I_2 + I_3 = I$$

$$\frac{v_1 - v_x}{R_1} + \frac{v_2 - v_x}{R_2} + \frac{v_3 - v_x}{R_3} = \frac{v_x - v_{out}}{R_F}$$

$$\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} \Rightarrow \frac{-v_{out}}{R_F}$$

$$v_{out} \Rightarrow -R_F \left\{ \frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} \right\}$$

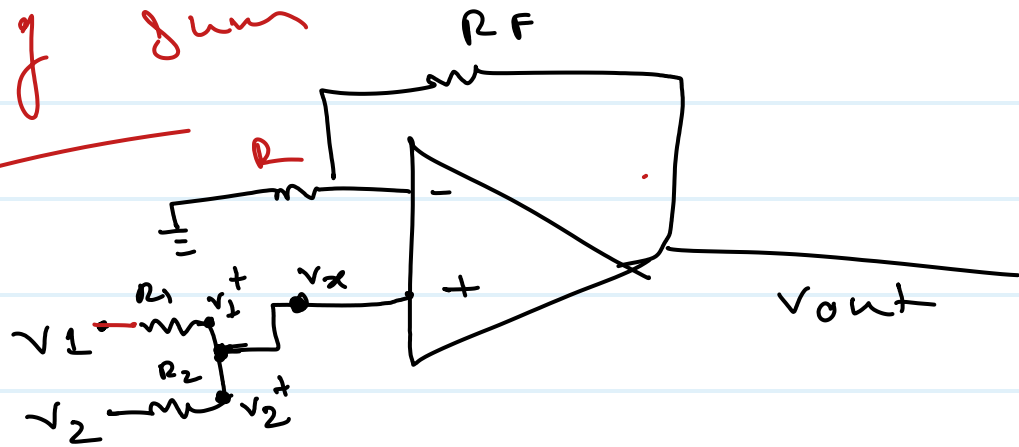
$$R_1 = R_2 = R_3 = R_F$$

$$v_{out} \Rightarrow -\frac{R}{R} \{v_1 + v_2 + v_3\}$$

$$v_{out} \Rightarrow (v_1 + v_2 + v_3) \times -1$$

$$\Rightarrow -(v_1 + v_2 + v_3)$$

non inventing sum



$$V_x \Rightarrow (V_1^+) + (V_2^+)$$

$$V_1^+ \Rightarrow \frac{R_2}{R_1 + R_2} \times V_1$$

$$V_x \Rightarrow \left(\frac{R_2}{R_1 + R_2} \right) V_1 + \left(\frac{R_1}{R_1 + R_2} \right) V_2$$

$$V_2^+ \Rightarrow \frac{R_1}{R_1 + R_2} \times V_2$$

$$A_{CL} \Rightarrow \frac{V_{out}}{V_x} \Rightarrow \left(1 + \frac{R_F}{R} \right) \frac{V_x}{V_x}$$

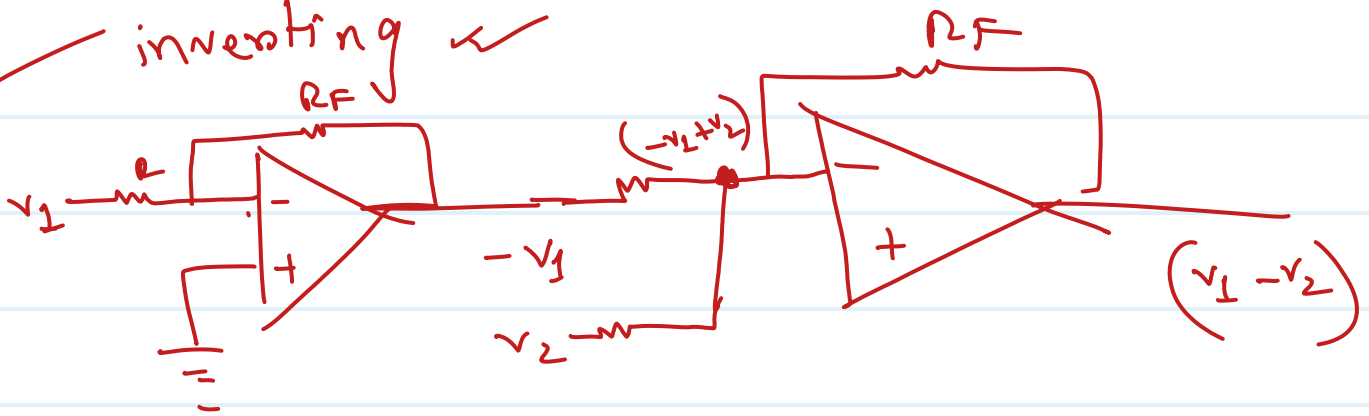
$$V_x \left(1 + \frac{R_F}{R} \right) \Rightarrow \left\{ \left(\frac{R_2}{R_1 + R_2} \right) V_1 + \left(\frac{R_1}{R_1 + R_2} \right) V_2 \right\} \left(1 + \frac{R_F}{R} \right)$$

$$\Rightarrow V_{out}$$

$$R_1 = R_2 = R = R_F \Rightarrow \left(\frac{R_2}{2R_2} V_1 + \frac{R_1}{2R_1} V_2 \right) (1 + 1)$$

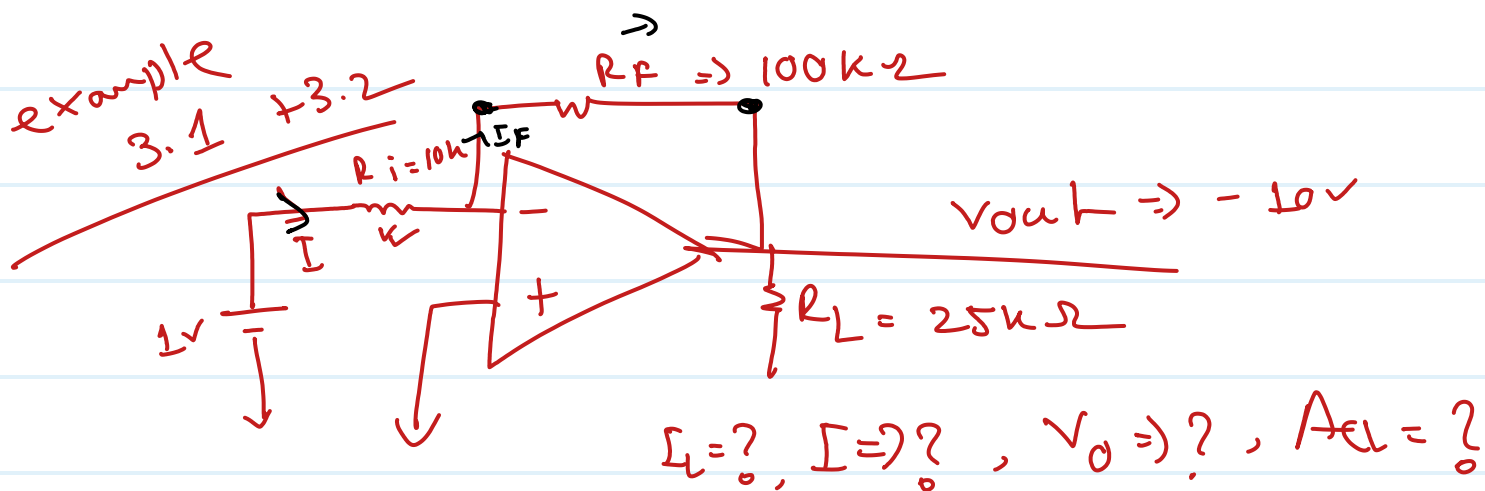
$$V_{out} \Rightarrow V_1 + V_2$$

Sub $\checkmark$ inverting $\checkmark$



$$\text{Sub} \Rightarrow (v_1 - v_2)$$

$$v_1 \approx 5 \\ v_2 \Rightarrow 3 \quad \text{Sub} \Rightarrow 5 - 3 \Rightarrow 2 \checkmark$$



$V \Rightarrow 1V , R_i \Rightarrow 10k\Omega$

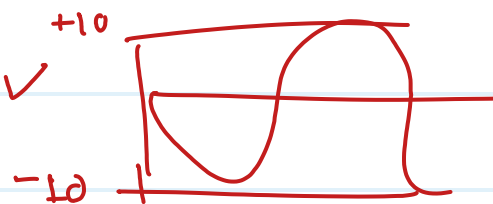
$$I \Rightarrow \frac{V}{R_i} = \frac{1V}{10k} \Rightarrow 0.1mA \quad \checkmark$$

$$\frac{V_{out}}{V_{in}} = \frac{-R_F}{R_i} = A_{cl}$$

$$V_{out} \Rightarrow - \frac{R_F}{R_i} \times V_{in} = - \frac{100k}{10k} \times 1$$

$$V_{out} \Rightarrow -10 \quad \checkmark$$

$V_{in} \Rightarrow 1V \quad \{ \quad V_{out} \Rightarrow -10V$



$$A_{cl} \Rightarrow \frac{-R_F}{R_i} \Rightarrow -100k$$

$$\frac{-10}{10k} = -10 \quad \checkmark$$

$$I_o \Rightarrow ?$$

$$I_L \Rightarrow \frac{v_{out}}{R_L} \Rightarrow \frac{10}{25k} \Rightarrow 0.4mA$$

$$I_o \Rightarrow I + I_L \Rightarrow 0.1mA + 0.4mA \\ \Rightarrow 0.5mA$$

voltage of R_F $\Rightarrow V_{R_F} \Rightarrow I \times R_F$

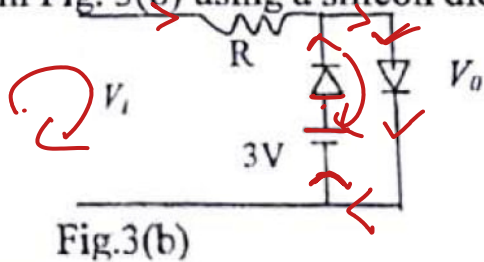
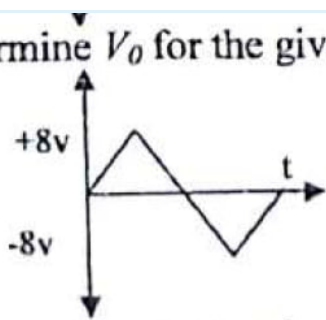
$$\Rightarrow 0.1 \times 100k \Omega$$

$$\Rightarrow \underline{\hspace{2cm}} \checkmark$$

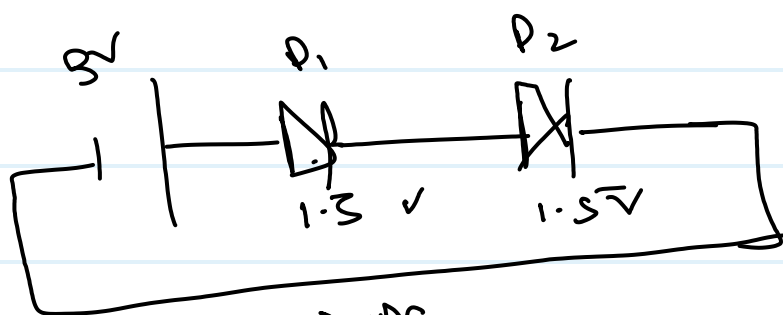
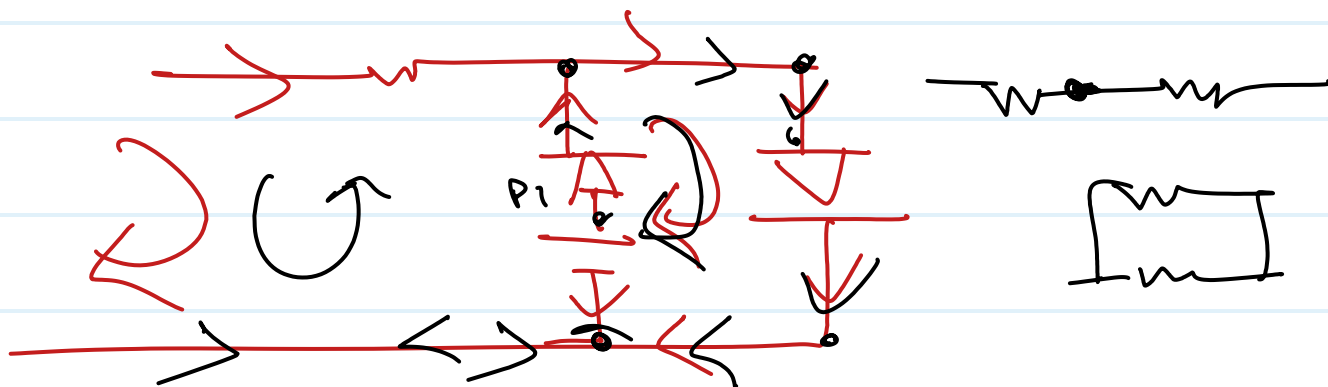
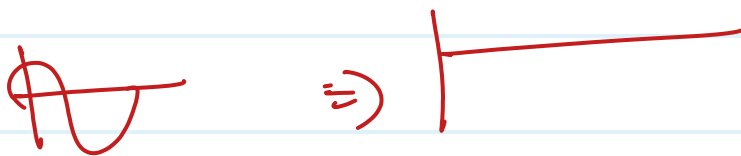
sum, sub

3.8, 3.9 example

1). Determine V_o for the given network in Fig. 3(b) using a silicon diode.



AC $\rightarrow$ DC



$$3 = D_1 + D_2$$

$$D_1 \Rightarrow \frac{3}{2}$$

$$\Rightarrow 1.5V$$

AC $\Rightarrow$ DC

